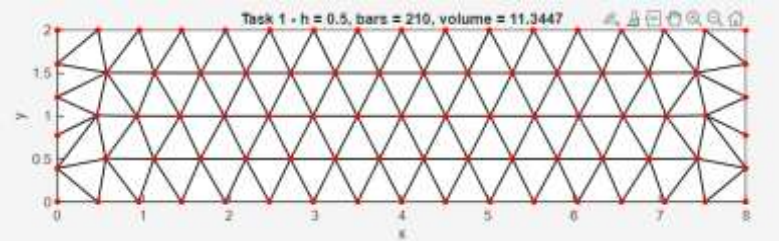
Structural Optimization 2026
Project #3 Report

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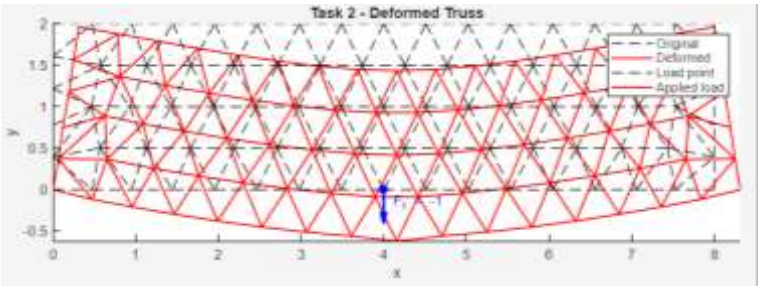
Task 1

For h = 0.5, the truss was generated using Distmesh with A = 0.1 for all bars. The structure contains 210 bars and has a volume of **11.3447**



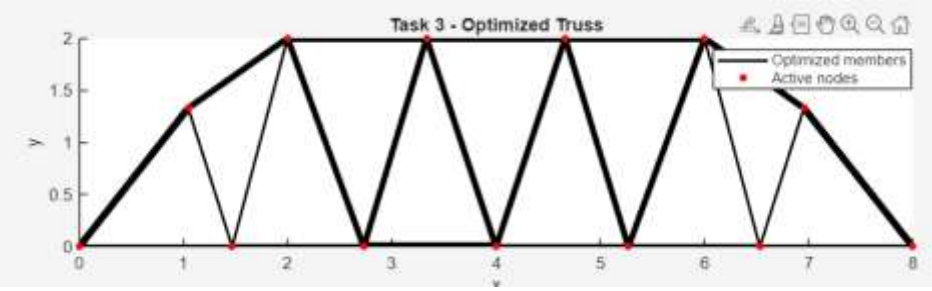
Task 2

For h = 0.5, the truss was loaded at (4,0) with Fx = 0 and Fy = -1. Using A = 0.1 and E = 500, the displacement at (4, 0) was calculated. The displacement obtained is ux at (4,0) is **0.031437**
and uy at (4,0) is **-0.12442** and the displacement magnitude **is 0.12833**



Task 3

For h = 1.4, the minimum-volume truss was optimized using linear programming. The stress constraint $|\sigma| \leq 1$ was imposed for the two load cases. The optimized volume obtained is **38.2187**



Task 4

The nested optimization was performed using GA for H and h, and linprog for the member areas. The best result was obtained for **H =3 and h = 4.5894 and the minimum volume found is 28**.

